3. Dopamine is converted to nor-epinephrine (Nor means No 'R' group, ie. Methyl group) by a reaction catalyzed by DOPA hydroxylase.
4. Nor-epinephrine is methylated to Epinephrine. The methyl group donor is S-adenosyl Methionine and the enzyme is N-methyl transferase.

Activity

1. Prepare a chart to explore the link between amino acid metabolism and TCA cycle
2. Prepare a chart to explain urea cycle.
3. With the help of e-books, try to prepare a list of syndromes associated with urea biosynthesis.

- Transaminases AST and ALT are important markers to assess liver damage. Many occupational hazards occur on exposure to solvents like CCl_4 and $CHCl_3$. **Example:** Liver damage.
- Insulin, Insulin like Growth factor-1 and Growth hormone have anabolic effects on protein metabolism.
- Ingestion of proteins or infusion with amino acids also have an anabolic effect on protein metabolism.

Summary

- Excess intake of amino acids are used for energy purposes, gluconeogenesis or ketogenesis, while their amino group is converted to urea.
- The carbon skeletons of amino acids are degraded on entry into TCA cycle.
- The first step in the catabolism of amino acids is the removal of its amino group. This removal can happen in two different ways - Transamination and Deamination.
- Transamination is a process, in which transfer of amino groups occur between a keto acid and an amino acid. Alanine transaminase (ALT) and Aspartate transaminase (AST) are the important transaminases.
- Aspartate transaminase catalyses the inter-conversion of aspartate and alpha ketoglutarate to oxaloacetate and glutamate, while alanine transaminase catalyses the conversion of alanine and alpha ketoglutarate to pyruvate and glutamate.

XII U4 Protein.indd 108 4/8/2020 3:09:25 PM